ShopNest_Week1

ShopNest - Online Shopping App

AUREX Full-Stack Engineering Internship - Month 3 | Week 1

Project Planning & Technical Design Document

1. Project Title

ShopNest - Full-Stack E-commerce Application

2. Problem Statement

Pakistan me choti shops aur local sellers ke paas apni online shop nahi hoti. Customers ko WhatsApp pe order karna parta hai, price aur stock ka pata nahi chalta. ShopNest is problem ko solve karega - ek complete website jahan seller apne products daale, customer browse karke cart me daale aur order kare.

3. Objectives

- User ko products browse, search aur filter karne ki facility dena

- Cart + Checkout + Order tracking system banana

- Admin ko products, orders aur users manage karne ka dashboard dena

- Secure Login/Register system banana (JWT + bcrypt)

4. Target Users & User Roles

1. Customer (User):

- Register/Login, Products dekhna, Search, Filter

- Cart me add karna, Order karna, Apne orders dekhna

2. Admin:

- Products Add/Edit/Delete

- Orders ka status change karna (Pending -> Shipped -> Delivered)

- Users dekhna, Dashboard dekhna

3. Guest:

- Bina login ke products dekh sakta hai, lekin order ke liye login lazmi hai

5. Core Features

- Login / Register (JWT + bcrypt)
- Product Listing with Category Filter + Search
- Product Detail Page
- Cart (Add / Remove / Quantity)
- Checkout + Order Creation
- Order History + Order Status
- Admin Dashboard
- Admin Product CRUD
- Admin Order Management

6. Functional Requirements

- User register kar sake, login kar sake
- Admin product add kar sake
- Customer cart me product add kar sake
- Stock se zyada order na ho sake
- Order ke baad stock automatically kam ho

7. Non-Functional Requirements

- Website mobile pe bhi responsive ho
- API ka response fast ho
- Passwords kabhi plain-text me save na hon
- .env me secrets secure hon

8. System Architecture

React Frontend (Vercel) -> REST API -> Node.js + Express.js Backend (Render) -> Mongoose -> MongoDB Atlas Database

Data Flow: User Form -> React -> API Request -> Express Route -> Controller -> Mongoose -> MongoDB -> Response -> React UI

9. Database Design

1. Users: _id, name, email, password (hashed), role (user/admin), createdAt

2. Products: _id, name, description, price, category, stock, imageUrl, createdBy (Admin), createdAt

3. Orders: _id, user (ref Users), items: [{ product (ref Products), quantity, price }], totalAmount, status (Pending/Shipped/Delivered/Cancelled), address, createdAt

4. Categories: _id, name, description

Relationships:

- User (1) -> Orders (Many)

- Order (Many) -> Products (Many)

- Category (1) -> Products (Many)

10. API List

Auth:

POST /api/auth/register

POST /api/auth/login

GET /api/auth/me

Products:

GET /api/products

GET /api/products/:id

POST /api/products - Admin Only

PUT /api/products/:id - Admin Only

DELETE /api/products/:id - Admin Only

Orders:

POST /api/orders - Order create karna

GET /api/orders/my - Apne orders dekhna

GET /api/orders - Admin Only

PUT /api/orders/:id/status - Admin Only

11. UI Flow

User Flow: Home -> Products -> Product Detail -> Cart -> Login -> Checkout -> Order Success -> My Orders

Admin Flow: Login -> Admin Dashboard -> Manage Products / Manage Orders

12. Technology Stack

Frontend: React, React Router, Axios, Tailwind CSS

Backend: Node.js, Express.js, Mongoose

Database: MongoDB Atlas

Auth: JWT, bcrypt

Tools: GitHub, Postman, Vercel, Render

13. GitHub Repository Structure

shopnest/

frontend/

backend/

README.md

.gitignore

14. Development Plan

Week 1: Planning (This Document)

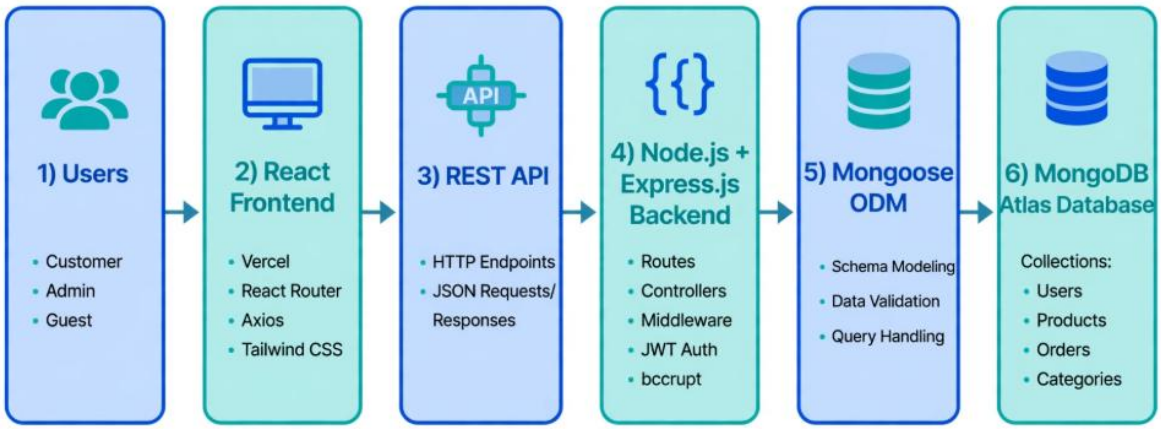
Week 2: Frontend + Backend + DB + APIs

Week 3: Auth + Cart/Order Logic + Testing

Week 4: Deployment + Docs + Presentation

ShopNest — Full-Stack E-Commerce App System Architecture

Data Flow: User Interactions → Frontend → API → Backend → ODM → Database



Technologies: React • Node.js • Express • MongoDB • Mongoose • JWT • REST